

The Chomsky Hierarchy

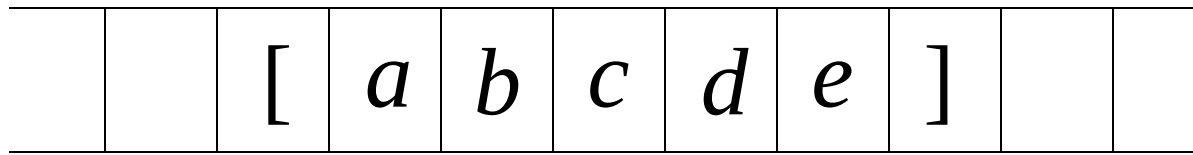
Linear-Bounded Automata:

Same as Non-Deterministic Turing Machines
with one difference:

the input string tape space
is the only tape space allowed to use

Linear Bounded Automaton (LBA)

Input string



Working space
in tape

Left-end
marker

Right-end
marker

All computation is done between end markers

We define LBA's as NonDeterministic

Open Problem:

NonDeterministic LBA's
have same power as
Deterministic LBA's ?

Example languages accepted by LBAs:

$$L = \{a^n b^n c^n\}$$

$$L = \{a^{n!}\}$$

LBA's have more power than PDA's
(pushdown automata)

LBA's have less power than Turing Machines

Unrestricted Grammars:

Productions

$$u \rightarrow v$$

String of variables
and terminals

String of variables
and terminals

Example unrestricted grammar:

$$S \rightarrow aBc$$

$$aB \rightarrow cA$$

$$Ac \rightarrow d$$

Theorem:

A language L is Turing-Acceptable
if and only if L is generated by an
unrestricted grammar

Context-Sensitive Grammars:

Productions

$$u \rightarrow v$$


String of variables
and terminals

String of variables
and terminals

and: $|u| \leq |v|$

The language $\{a^n b^n c^n\}$

is context-sensitive:

$$S \rightarrow abc \mid aAbc$$

$$Ab \rightarrow bA$$

$$Ac \rightarrow Bbcc$$

$$bB \rightarrow Bb$$

$$aB \rightarrow aa \mid aaA$$

Theorem:

A language L is context sensitive
if and only if
it is accepted by a Linear-Bounded automaton

Observation:

There is a language which is decidable but not
context sensitive

The Chomsky Hierarchy

Non Turing-Acceptable

Turing-Acceptable

decidable

Context-sensitive

Context-free

Regular